실습과제 1

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Nov  2 16:23:27 KST 2026

  System load:  0.56               Processes:             61
  Usage of /:   0.6% of 1006.85GB  Users logged in:       0
  Memory usage: 5%                 IPv4 address for eth0: 172.28.237.8
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

This message is shown once a day. To disable it please create the
/home/linux/.hushlogin file.
linux@BOOK-M8LSJ7JPPD: $ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD: $ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD: $ cd ~/ros2_ws/src
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ ls
first_pkg  pub1  pub1_1  pub1_2  pub1_3  second_pkg
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ ros2 pkg create pub2_1 --build-type ament_cmake --dependencies rclcpp std_msgs
going to create a new package
package name: pub2_1
destination directory: /home/linux/ros2_ws/src
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['linux <linux@todo.todo>']
licenses: ['TODO: License declaration']
build type: ament_cmake
dependencies: ['rclcpp', 'std_msgs']
creating folder ./pub2_1
creating ./pub2_1/package.xml
creating source and include folder
creating folder ./pub2_1/src
creating folder ./pub2_1/include/pub2_1
creating ./pub2_1/CMakeLists.txt

[WARNING]: Unknown license 'TODO: License declaration'.  This has been set in the package.xml, but no LICENSE file has been created.
It is recommended to use one of the ament license identifiers:
Apache-2.0
BSL-1.0
BSD-2.0
BSD-2-Clause
BSD-3-Clause
GPL-3.0-only
LGPL-3.0-only
MIT
MIT-0
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ tree pub2_1
pub2_1
├── CMakeLists.txt
├── include
│   └── pub2_1
├── package.xml
└── src

4 directories, 2 files
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ nano pub2_1/src/pub.cpp
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
 GNU nano 7.2                                  pub2_1/src/pub.cpp
#include "rclcpp/rclcpp.hpp"
#include "std_msgs/msg/int32.hpp"
#include <memory>
#include <chrono>
#include <functional>

using namespace std::chrono_literals;

// 타이머가 호출하는 콜백함수
void callback(
    rclcpp::Node::SharedPtr node,
    rclcpp::Publisher<std_msgs::msg::Int32>::SharedPtr mypub)
{
    static int count = 0;

    // Int32 메시지 생성
    auto message = std_msgs::msg::Int32();

    // 0부터 시작해서 1씩 증가
    message.data = count++;

    // 발행 내용 출력
    RCLCPP_INFO(node->get_logger(),
                "Publish: %d", message.data);

    // 메시지 발행
    mypub->publish(message);
}

int main(int argc, char* argv[])
{
    // ROS2 초기화
    rclcpp::init(argc, argv);

    // 노드 생성
    auto node = std::make_shared<rclcpp::Node>("mynode");
                                [ Read 58 lines ]
^C Help         ^O Write Out    ^W Where Is     ^K Cut          ^T Execute      ^C Location     M-U Undo        M-A Set Mark    M-] To Bracket
^X Exit         ^R Read File    ^\ Replace      ^U Paste        ^J Justify      ^/ Go To Line   M-E Redo        M-6 Copy        ^Q Where Was
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ tree pub2_1
pub2_1
├── CMakeLists.txt
├── include
│   └── pub2_1
├── package.xml
└── src
    └── pub.cpp

4 directories, 3 files
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ nano pub2_1/CMakeLists.txt
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
 GNU nano 7.2                                pub2_1/CMakeLists.txt
cmake_minimum_required(VERSION 3.16)
project(pub2_1)

# Default to C++14
if(NOT CMAKE_CXX_STANDARD)
    set(CMAKE_CXX_STANDARD 14)
endif()

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")
    add_compile_options(-Wall -Wextra -Wpedantic)
endif()

# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(std_msgs REQUIRED)

# build
add_executable(pub2_1 src/pub.cpp)
ament_target_dependencies(pub2_1 rclcpp std_msgs)

# Install
install(TARGETS
    pub2_1
    DESTINATION lib/${PROJECT_NAME}
)

ament_package()
                                [ Read 28 lines ]
^C Help         ^O Write Out    ^W Where Is     ^K Cut          ^T Execute      ^C Location     M-U Undo        M-A Set Mark    M-] To Bracket
^X Exit         ^R Read File    ^\ Replace      ^U Paste        ^J Justify      ^/ Go To Line   M-E Redo        M-6 Copy        ^Q Where Was
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cat pub2_1/CMakeLists.txt
cmake_minimum_required(VERSION 3.16)
project(pub2_1)

# Default to C++14
if(NOT CMAKE_CXX_STANDARD)
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endif()

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# find dependencies
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find_package(rclcpp REQUIRED)
find_package(std_msgs REQUIRED)

# build
add_executable(pub2_1 src/pub.cpp)
ament_target_dependencies(pub2_1 rclcpp std_msgs)

# Install
install(TARGETS
    pub2_1
    DESTINATION lib/${PROJECT_NAME}
)

ament_package()
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --symlink-install --packages-select pub2_1
Starting >>> pub2_1
Finished <<< pub2_1 [14.0s]

Summary: 1 package finished [14.4s]
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ source install/local_setup.bash
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ ros2 run pub2_1 pub2_1
[INFO] [1793604624.540905816] [mynode]: Publish: 0
[INFO] [1793604624.641061133] [mynode]: Publish: 1
[INFO] [1793604624.740912041] [mynode]: Publish: 2
[INFO] [1793604624.841153950] [mynode]: Publish: 3
[INFO] [1793604624.941129813] [mynode]: Publish: 4
[INFO] [1793604625.041185325] [mynode]: Publish: 5
[INFO] [1793604625.141086230] [mynode]: Publish: 6
[INFO] [1793604625.240943291] [mynode]: Publish: 7
[INFO] [1793604625.340965026] [mynode]: Publish: 8
[INFO] [1793604625.441045021] [mynode]: Publish: 9
[INFO] [1793604625.540918821] [mynode]: Publish: 10
[INFO] [1793604625.641058245] [mynode]: Publish: 11
[INFO] [1793604625.741028348] [mynode]: Publish: 12
[INFO] [1793604625.840993223] [mynode]: Publish: 13
[INFO] [1793604625.940914648] [mynode]: Publish: 14
[INFO] [1793604626.041012681] [mynode]: Publish: 15
[INFO] [1793604626.140932273] [mynode]: Publish: 16
[INFO] [1793604626.241116441] [mynode]: Publish: 17
```

```
linux@BOOK-M8LSJ7JPPD:~$ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD:~$ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD:~$ ros2 node list
/mynode
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic list -t
/mytopic [std_msgs/msg/Int32]
/parameter_events [rcl_interfaces/msg/ParameterEvent]
/rosout [rcl_interfaces/msg/Log]
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic echo /mytopic
data: 786
---
data: 787
---
data: 788
---
data: 789
---
data: 790
---
data: 791
---
data: 792
---
data: 793
---
data: 794
---
data: 795
---
data: 796
---
data: 797
---
data: 798
---
data: 799
---
data: 800
---
data: 801
```

```
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic hz /mytopic
average rate: 10.002
        min: 0.099s max: 0.101s std dev: 0.00058s window: 12
average rate: 9.998
        min: 0.099s max: 0.101s std dev: 0.00044s window: 22
average rate: 9.999
        min: 0.099s max: 0.101s std dev: 0.00043s window: 33
average rate: 9.996
        min: 0.099s max: 0.101s std dev: 0.00043s window: 43
average rate: 9.999
        min: 0.099s max: 0.101s std dev: 0.00047s window: 54
average rate: 9.999
        min: 0.099s max: 0.101s std dev: 0.00047s window: 64
average rate: 10.000
        min: 0.099s max: 0.101s std dev: 0.00046s window: 75
average rate: 10.000
        min: 0.099s max: 0.101s std dev: 0.00047s window: 86
average rate: 10.000
        min: 0.099s max: 0.101s std dev: 0.00044s window: 97
average rate: 10.000
        min: 0.098s max: 0.102s std dev: 0.00053s window: 108
average rate: 10.000
        min: 0.098s max: 0.102s std dev: 0.00052s window: 118
average rate: 9.999
        min: 0.098s max: 0.102s std dev: 0.00050s window: 129
average rate: 10.000
        min: 0.098s max: 0.102s std dev: 0.00049s window: 140
average rate: 10.000
        min: 0.098s max: 0.102s std dev: 0.00048s window: 150
average rate: 10.000
        min: 0.098s max: 0.102s std dev: 0.00048s window: 160
average rate: 10.000
```

```
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic bw /mytopic
Subscribed to [/mytopic]
86 B/s from 10 messages
        Message size mean: 8 B min: 8 B max: 8 B
83 B/s from 20 messages
        Message size mean: 8 B min: 8 B max: 8 B
82 B/s from 30 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 40 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 50 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 60 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 70 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 80 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 90 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 100 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 100 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 100 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 100 messages
        Message size mean: 8 B min: 8 B max: 8 B
81 B/s from 100 messages
```

```
^Clinux@BOOK-M8LSJ7JPPD:~$ rqt_graph
```

